Simple Yet Efficient Algorithms for Maximum Inner Product Search via Extreme Order Statistics

Ninh Pham

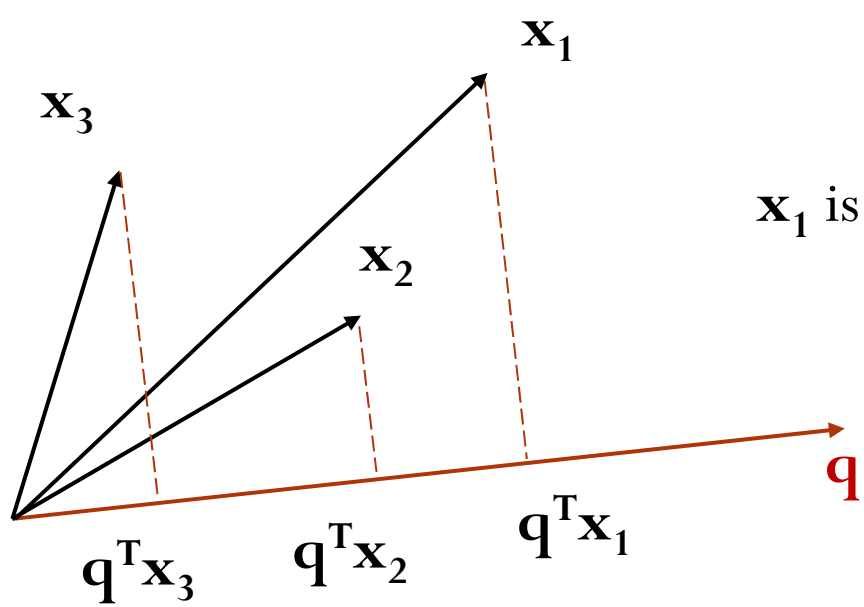
University of Auckland

KDD, Aug 21, 2021

Maximum inner product search

- Maximum inner product search (MIPS):
  - Given a data set $\mathbf{X} \subset \mathbf{R}^{\mathbf{d}\times\mathbf{n}}$ of size $\mathbf{n}$ and a query $\mathbf{q} \in \mathbf{R}^{\mathbf{d}}$, return the point $\mathbf{x} \in \mathbf{X}$ such that the inner product $\mathbf{q}^{\mathbf{T}}\mathbf{x}$ is maximum.

- Geometric intuition:



$\mathbf{x}_1$ is the top-1 MIPS.

Challenges of MIPS

- Problem:
  - Given a data structure built in $\tilde{O}(dn)$ time and space, can we have a fast MIPS solver that returns the best search recall?
- Drawbacks of prior arts:
  - Linear scan: The SETH conjecture is wrong if existing a truly sublinear time exact MIPS solver [ARW17].
  - Locality-sensitive hashing (LSH): Subquadratic indexing time and space to guarantee approximate MIPS in sublinear time [SL14].
  - Delaunay graph: Quadratic time for constructing the index [MB18].

Contribution: CEOs as MIPS solvers

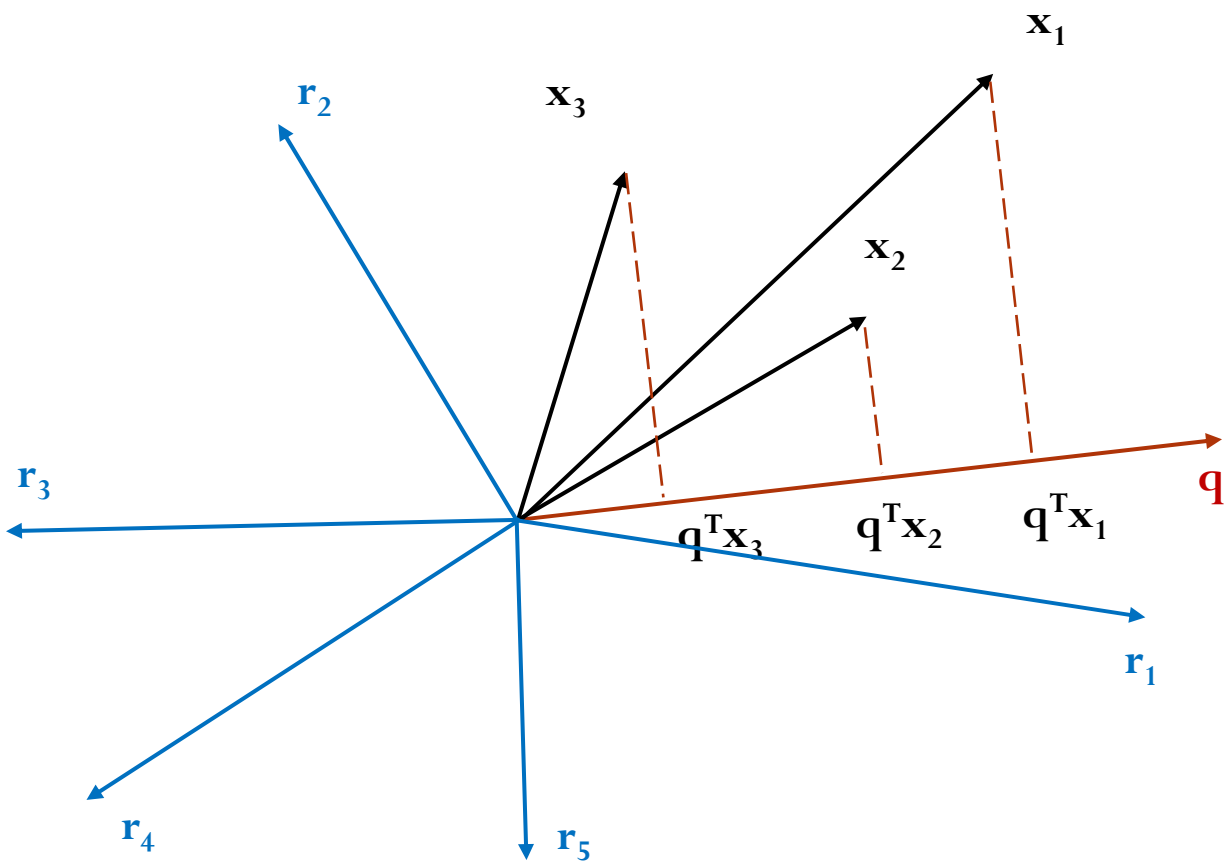
- In theory, CEOs is a novel dimensionality reduction based on the theory of Concomitant of Extreme Order statistics.
  - Has sublinear query time for exact MIPS with search recall guarantees under a mild condition.
  - Is an optimal dimensionality reduction on a unit sphere (i.e. cosine similarity).

Contribution: CEOs as MIPS solvers

- In practice, CEOs variants need $\tilde{\mathbf{O}}(\mathbf{dn})$ indexing time.
  - CEOs-Est: A fast dimensionality reduction method.
  - CEOs-TA: A fast threshold algorithm (TA) based top-k MIPS solver.
  - coCEOs: A budgeted MIPS solver runs in $\mathbf{O(B)}$ given $\mathbf{B}$ budget computations.
- Empirically, CEOs variants are simple and efficient.
  - Build indexes in just minutes in a single machine.
  - Achieve competitive search recall compared to recent MIPS solvers.

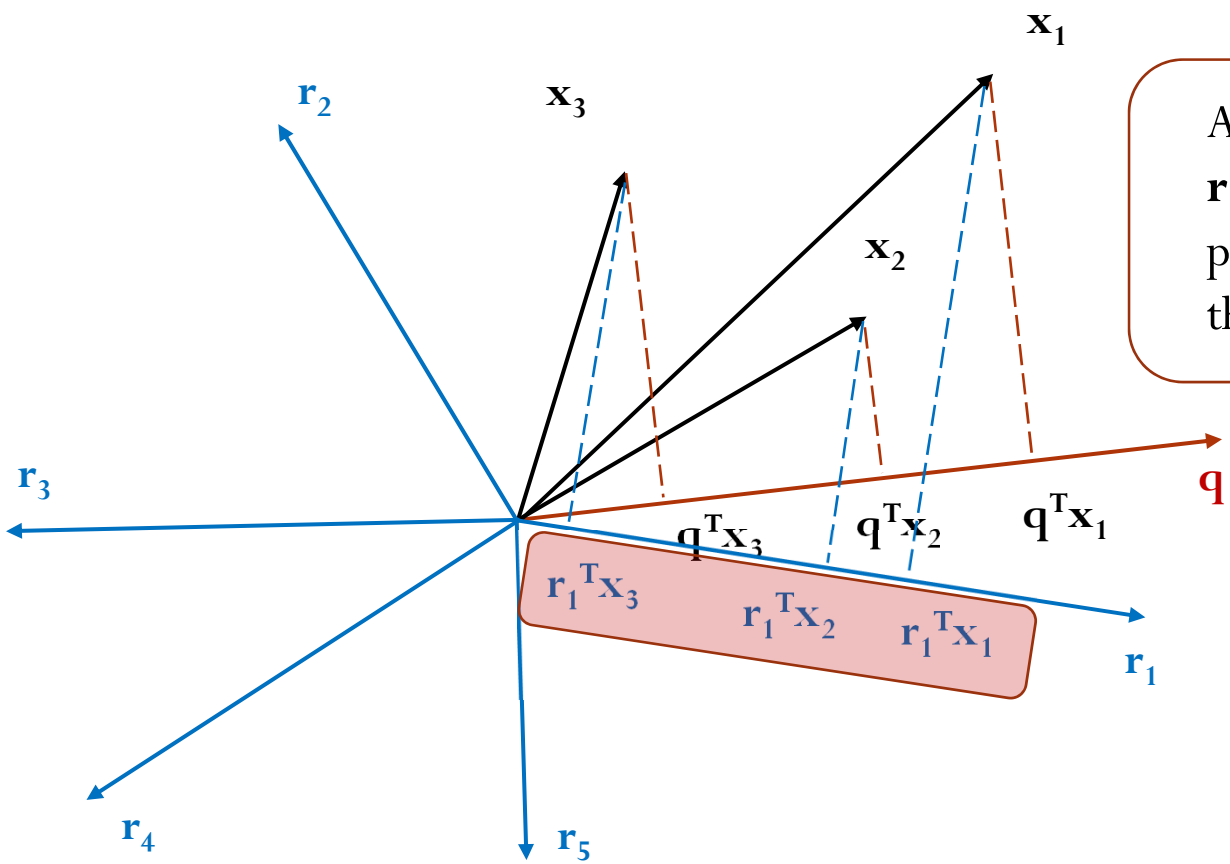
CEOs: Geometric intuition

- CEOs utilizes a large number of random projections.



CEOs: Geometric intuition

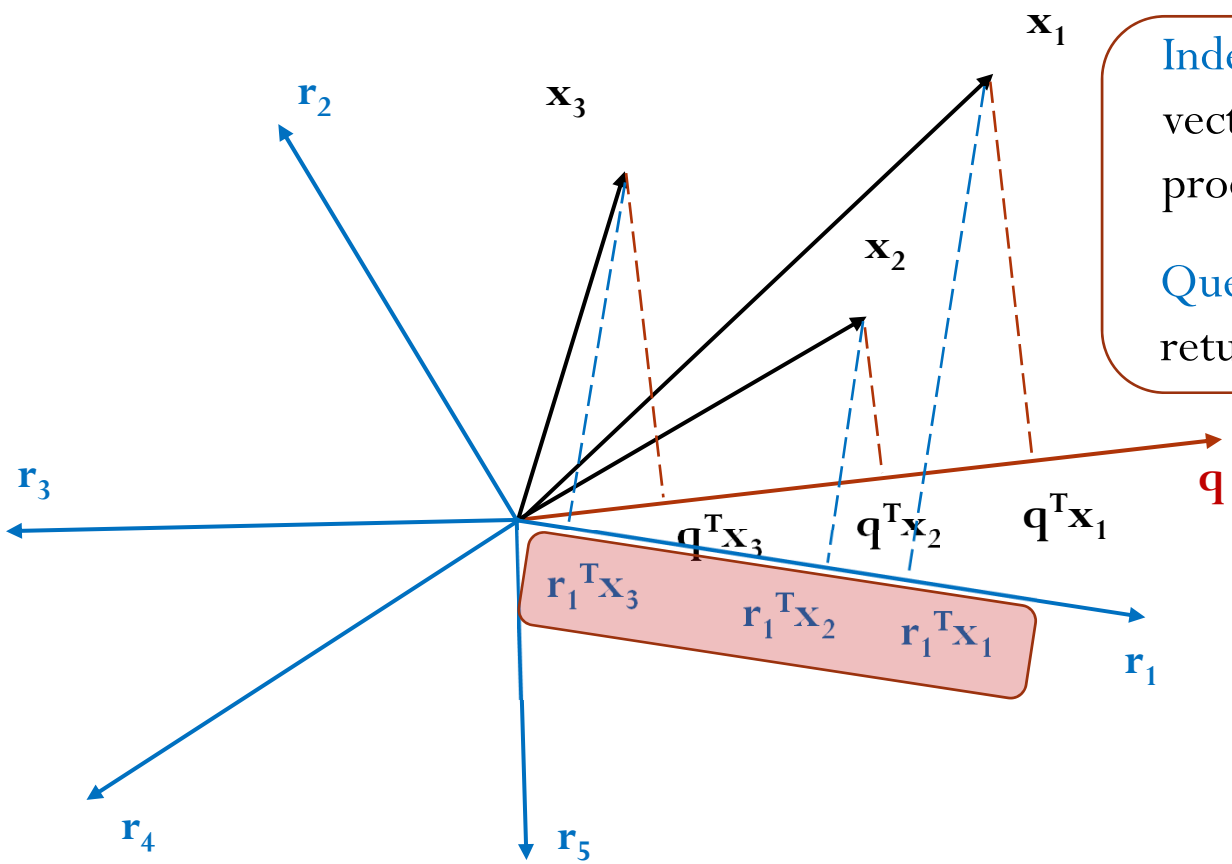
- CEOs utilizes a large number of random projections.



Among **5** random vectors $\mathbf{r}_i$, $\mathbf{r}_1$ is closest to $\mathbf{q}$ and the projections into $\mathbf{r}_1$ preserves the inner product order.

CEOs: Geometric intuition

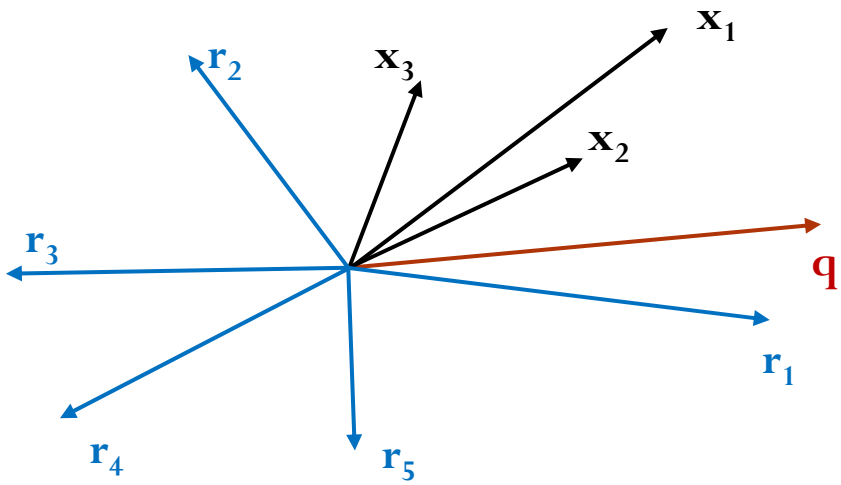
- CEOs utilizes a large number of random projections.



Indexing: Generate many random vectors $\mathbf{r}_i$, precompute the inner product order $\mathbf{L}_i$ for $\mathbf{r}_i$

Querying: Find $\mathbf{r}_i$ closest to $\mathbf{q}$ and return the top-k MIPS from $\mathbf{L}_i$

CEOs: Dimensionality reduction



Among **5** random vectors $\mathbf{r}_i$, we only use $\mathbf{r}_1$ to estimate inner products.

				$\mathbf{r}_1$	$\mathbf{r}_2$	$\mathbf{r}_3$	$\mathbf{r}_4$	$\mathbf{r}_5$
$\mathbf{x}_1$		1.2	0.9	4	-2	-4	-3	-3
$\mathbf{x}_2$		0.5	0.5	3	-1	-3	-4	-4
$\mathbf{x}_3$		0.1	0.7	1	-3	-2	-3	-3
$\mathbf{q}$		1.5	0.1	9	4	-8	-3	0

Concomitants of Extreme Order statistics

Random projections: Using **D** random vectors $\mathbf{r_i}$ in $\mathbf{N(0, 1)}$, we have **D** bivariate samples $\mathbf{(Q_i, X_i)}$ from $\mathbf{N(0, 0, 1, \| x \|, x^T q)}$ where $\mathbf{Q_i = q^T r_i}$ and $\mathbf{X_i = x^T r_i}$

x

1.2	0.9

$\mathbf{X_i = x^T r_i}$ ⇨

5	-2	4	-3	-3

q

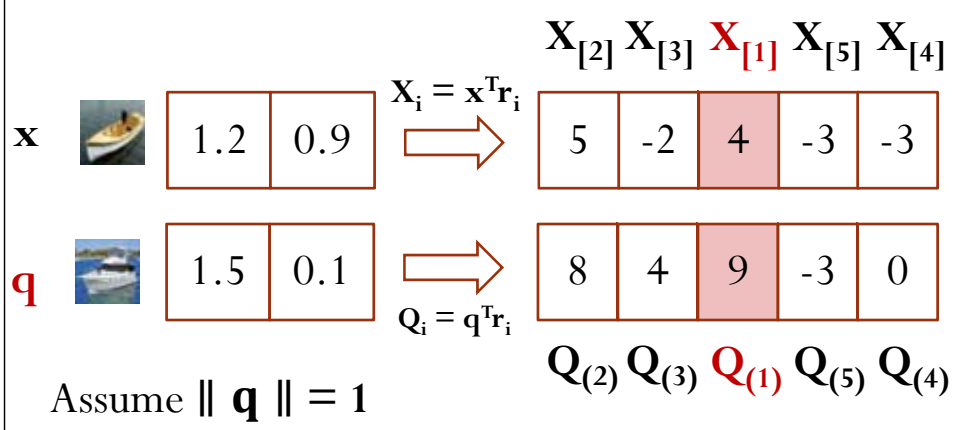
1.5	0.1

⇨ $\mathbf{Q_i = q^T r_i}$

8	4	9	-3	0

Assume $\| \mathbf{q} \| = 1$

Concomitants of Extreme Order statistics



Assume $\| \mathbf{q} \| = 1$

Random projections: Using **D** random vectors $\mathbf{r}_i$ in $\mathbf{N(0, 1)}$, we have **D** bivariate samples $(\mathbf{Q_i}, \mathbf{X_i})$ from $\mathbf{N(0, 0, 1, \| x \|, x^T q)}$ where $\mathbf{Q_i = q^T r_i}$ and $\mathbf{X_i = x^T r_i}$

Order statistics: Sort **D** pairs $(\mathbf{Q_i}, \mathbf{X_i})$ by **Q**-value, we form the order statistics where $\mathbf{Q_{(1)}}$ is the first order statistics and $\mathbf{X_{[1]}}$ is the concomitant of the first order statistics.

Concomitants of Extreme Order statistics

Concomitants of Extreme Order statistics

$\Downarrow$

					$X_{[2]}$	$X_{[3]}$	$X_{[1]}$	$X_{[5]}$	$X_{[4]}$
$\mathbf{x}$		1.2	0.9	$X_i = x^T r_i$ $\Rightarrow$	5	-2	4	-3	-3
$\mathbf{q}$		1.5	0.1	$\Rightarrow$ $Q_i = q^T r_i$	8	4	9	-3	0
					$Q_{(2)}$	$Q_{(3)}$	$Q_{(1)}$	$Q_{(5)}$	$Q_{(4)}$

Assume $\| \mathbf{q} \| = 1$

$\Uparrow$

Extreme Order statistics

Random projections: Using **D** random vectors $\mathbf{r_i}$ in $\mathbf{N(0, 1)}$, we have **D** bivariate samples $\mathbf{(Q_i, X_i)}$ from $\mathbf{N(0, 0, 1, \| x \|, x^T q)}$ where $\mathbf{Q_i = q^T r_i}$ and $\mathbf{X_i = x^T r_i}$

Order statistics: Sort **D** pairs $\mathbf{(Q_i, X_i)}$ by **Q**-value, we form the order statistics where $\mathbf{Q_{(1)}}$ is the first order statistics and $\mathbf{X_{[1]}}$ is the concomitant of the first order statistics.

Extreme order statistics: When **D** is sufficiently large, $\mathbf{Q_{(1)}}$ is the extreme order statistics and $\mathbf{X_{[1]}}$ is the concomitant of the extreme order statistics.

Theory of Concomitants of Extreme Order statistics [DG74]

Concomitants of Extreme Order statistics

$\Downarrow$

				$X_{[2]}$	$X_{[3]}$	$X_{[1]}$	$X_{[5]}$	$X_{[4]}$
$\mathbf{x}$		1.2	0.9	$X_i = x^T r_i$ $\Rightarrow$	5	-2	4	-3	-3
$\mathbf{q}$		1.5	0.1	$\Rightarrow$ $Q_i = q^T r_i$	8	4	9	-3	0
					$Q_{(2)}$	$Q_{(3)}$	$Q_{(1)}$	$Q_{(5)}$	$Q_{(4)}$

Assume $\| \mathbf{q} \| = 1$

$\Uparrow$

Extreme Order statistics

Extreme order statistics: $Q_{(1)}$ is the maximum variable among $\mathbf{D}$ random variables $Q_i = q^T r_i \sim N(0, 1)$.

Extreme order statistics:

$$E[Q_{(1)}] \rightarrow \sqrt{2\log(D)},\ Var[Q_{(1)}] \rightarrow 0$$

Concomitant of extreme order statistics:

$$X_{[1]} \rightarrow N\left(x^T q \sqrt{2\log(D)},\ \| x \|^2 - (x^T q)^2\right)$$

Theory of Concomitants of Extreme Order statistics [DG74]

Concomitants of
Extreme Order statistics

				$X_{[D-1]}$		$X_{[1]}$		$X_{[2]}$	$X_{[D]}$
$\mathbf{x}$		1.2	0.9	$X_i = x^T r_i$ ⇒	-5	-2	4	• • •	3	-8
$\mathbf{q}$		1.5	0.1	$Q_i = q^T r_i$ ⇒	-8	4	9	• • •	7	-9
				$Q_{(D-1)}$		$Q_{(1)}$		$Q_{(2)}$	$Q_{(D)}$

Assume $\| \mathbf{q} \| = 1$

Extreme Order statistics

Top s_0 maximum and minimum order statistics: $Q_{(i)}$ and $Q_{(D-i+1)}$ where $i = 1, \ldots, s_0$

Extreme order statistics:

$$E[Q_{(i)}] \rightarrow \sqrt{2\log(D)}$$

$$E[Q_{(D-i+1)}] \rightarrow -\sqrt{2\log(D)}$$

Concomitant of extreme order statistics:

$$X_{[i]} \rightarrow N(x^T q \sqrt{2\log(D)},\ \| x \|^2 - (x^T q)^2)$$

$$X_{[D-i+1]} \rightarrow N(-x^T q \sqrt{2\log(D)},\ \| x \|^2 - (x^T q)^2)$$

$X_{[i]}$ and $X_{[D-i+1]}$ are independent asymptotically.

CEOs: Estimate inner product

Concomitants of
Extreme Order statistics

				$X_{[D-1]}$		$X_{[1]}$		$X_{[2]}$	$X_{[D]}$
$\mathbf{x}$	1.2	0.9	$X_i = x^T r_i$ ⇒	-5	-2	4	• • •	3	-8
$\mathbf{q}$	1.5	0.1	⇒ $Q_i = q^T r_i$	-8	4	9	• • •	7	-9
				$Q_{(D-1)}$		$Q_{(1)}$		$Q_{(2)}$	$Q_{(D)}$

Assume $\| \mathbf{q} \| = 1$

Extreme Order statistics

Top s_0 maximum and minimum order statistics: $Q_{(i)}$ and $Q_{(D-i+1)}$ where $i = 1, \ldots, s_0$

Concomitant of extreme order statistics:

$$X_{[i]} \rightarrow N\left(x^T q \sqrt{2\log(D)},\ \| x \|^2 - (x^T q)^2\right)$$

$$X_{[D-i+1]} \rightarrow N\left(-x^T q \sqrt{2\log(D)},\ \| x \|^2 - (x^T q)^2\right)$$

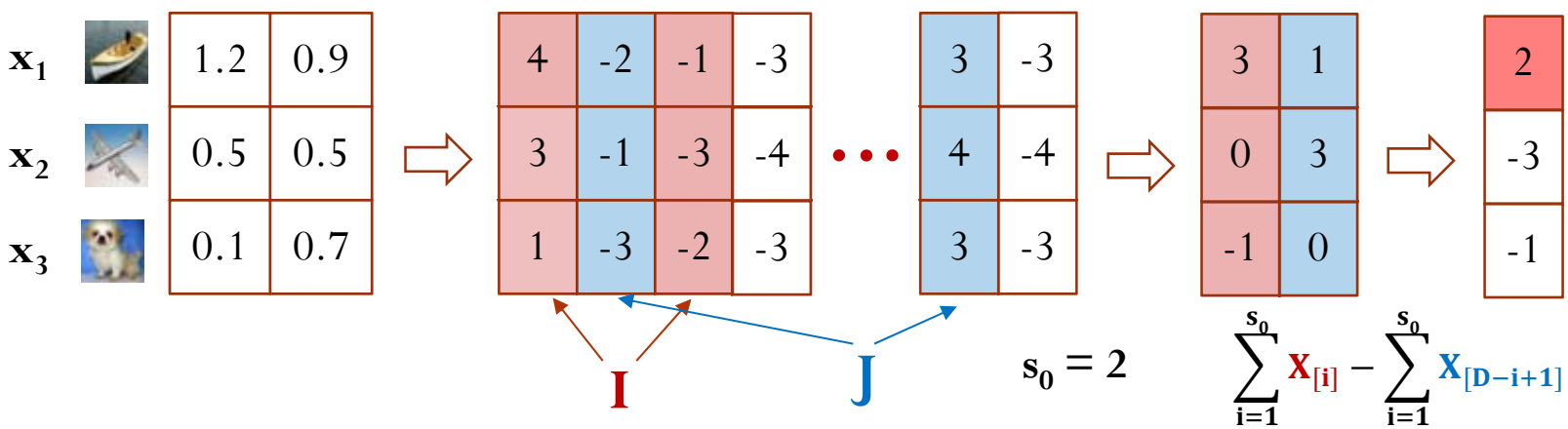
$X_{[i]}$ and $X_{[D-i+1]}$ are independent asymptotically.

Estimate inner product $x^T q$:

$$\sum_{i=1}^{s_0} X_{[i]} - \sum_{i=1}^{s_0} X_{[D-i+1]}$$

CEOs: Sublinear MIPS solver

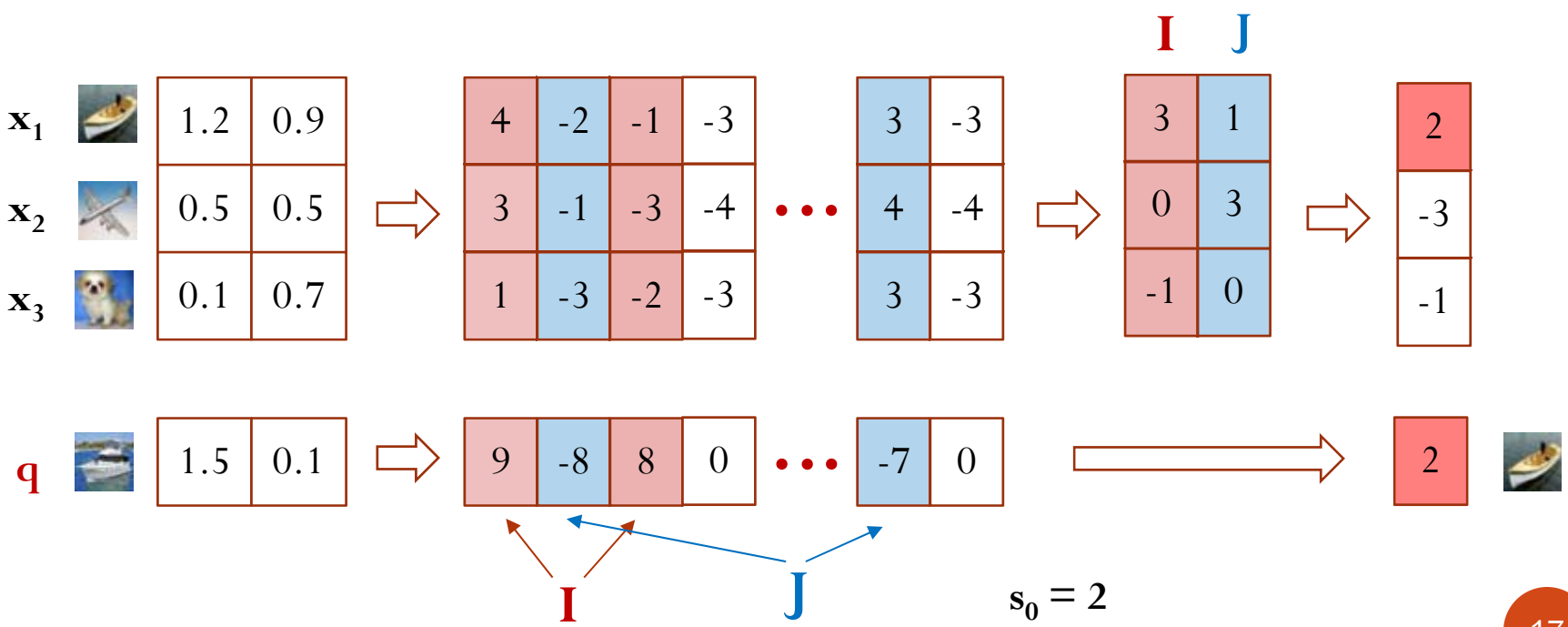
- Observation:
  - Given **D** random projections, any subset of $\mathbf{s_0}$ distinct indexes in **[D]** can be the concomitants of $\mathbf{s_0}$ maximum or minimum order statistics.
  - We can precompute the estimators of all **n** inner products for any query.
- Indexing:
  - Execute **D** random projections on **X**.
  - Precompute top-k MIPS for $\mathbf{O(D/s_0)^{s_0}}$ pairs **(I, J)** where **I** and **J** are subsets of $\mathbf{s_0}$ distinct indexes in **[D]**.



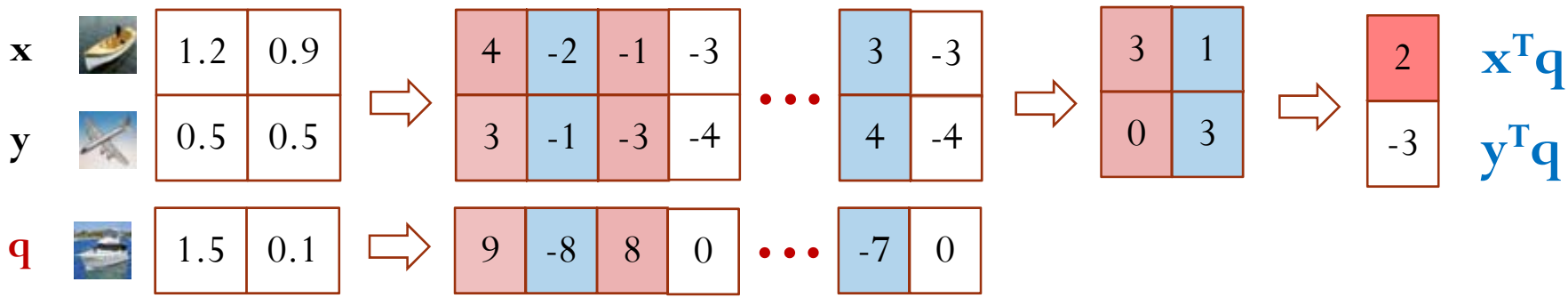
$$\sum_{i=1}^{s_0} X_{[i]} - \sum_{i=1}^{s_0} X_{[D-i+1]}$$

CEOs: Sublinear MIPS solver

- Querying:
  - Execute **D** random projection on **q**.
  - Compute two subsets of $\mathbf{s_0}$ indexes: **I** for maximum and **J** for minimum values
  - Return the top-k MIPS based on **I** and **J**



CEOs: Sublinear MIPS solver



Concomitant of extreme order statistics:

$$X_{[i]} \rightarrow N\left(x^Tq\sqrt{2\log(D)},\ \|x\|^2-(x^Tq)^2\right)$$

$$Y_{[i]} \rightarrow N\left(y^Tq\sqrt{2\log(D)},\ \|y\|^2-(y^Tq)^2\right)$$

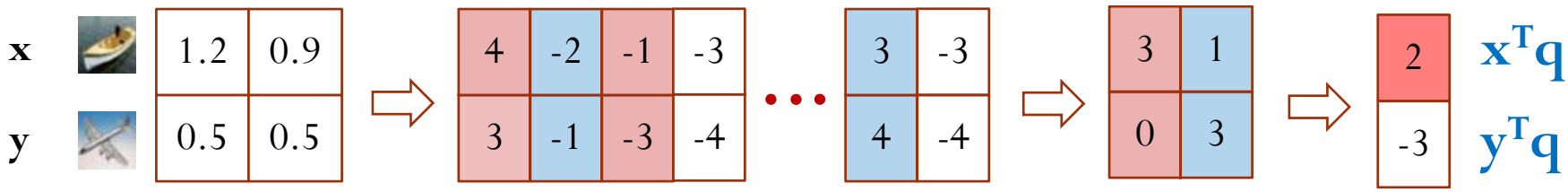
Estimate inner product x^Tq:

$$\sum_{i=1}^{s_0} X_{[i]} - \sum_{i=1}^{s_0} X_{[D-i+1]}$$

Estimate inner product y^Tq:

$$\sum_{i=1}^{s_0} Y_{[i]} - \sum_{i=1}^{s_0} Y_{[D-i+1]}$$

CEOs: Sublinear MIPS solver



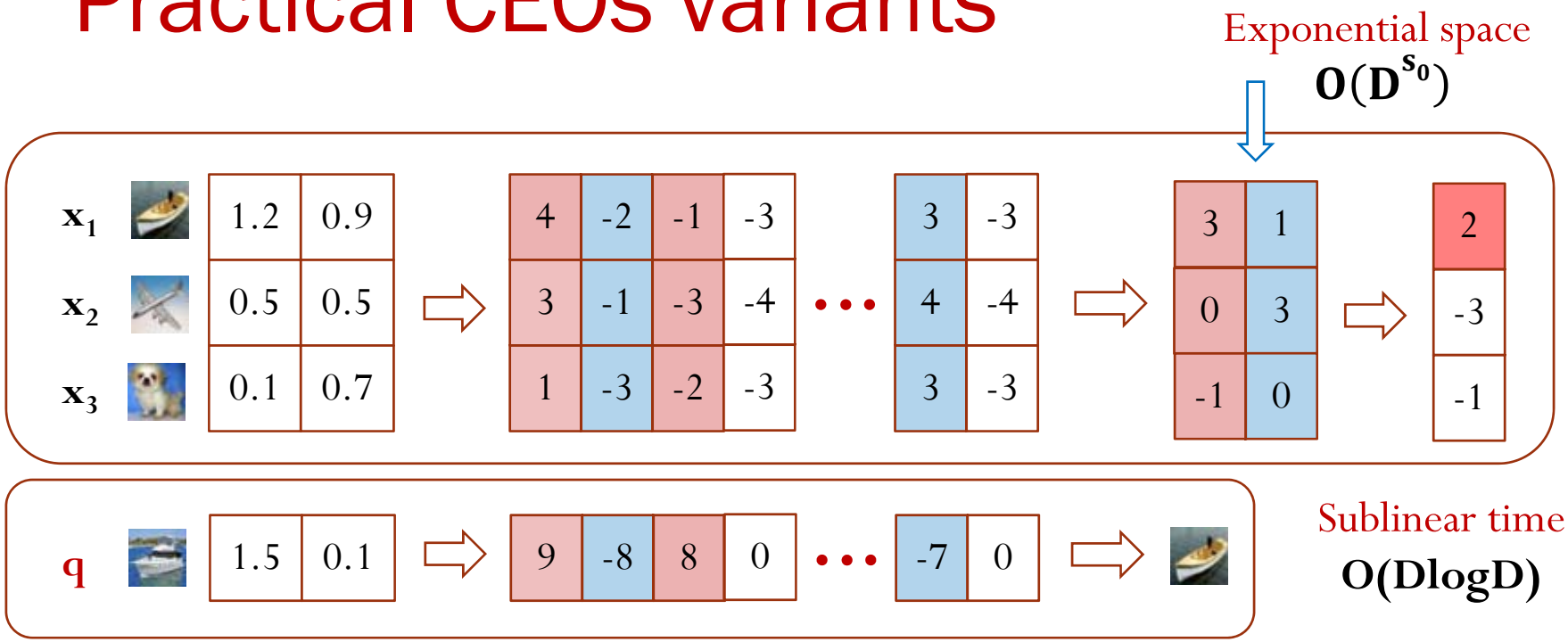
- Main theorem:
  - Let $\mathbf{x} \in \mathbf{X}$ be the top-1 MIPS for $\mathbf{q}$. Let $2\mathbf{s_0}$ be the number of concomitants.
  - Let $\boldsymbol{\alpha}_\mathbf{y} = (\mathbf{x}^T\mathbf{q} - \mathbf{y}^T\mathbf{q})/\left(\sqrt{\|\mathbf{x}\|^2 - (\mathbf{x}^T\mathbf{q})^2} + \sqrt{\|\mathbf{y}\|^2 - (\mathbf{y}^T\mathbf{q})^2}\right)$.
  - Assume that $\boldsymbol{\alpha}_\mathbf{y}^2 \geq \mathbf{c}/\mathbf{s_0}$ for all $\mathbf{y} \in \mathbf{X}\backslash\mathbf{x}$ and $\mathbf{c} > 1$. By choosing $\mathbf{D} = \mathbf{n}^{1/c}$, for all $\mathbf{y} \in \mathbf{X}\backslash\mathbf{x}$ we have:

$$\mathbf{Pr}\left[\sum_{r=1}^{s_0}\left(X_{[r]} - X_{[D-r+1]}\right) \geq \sum_{r=1}^{s_0}\left(Y_{[r]} - Y_{[D-r+1]}\right)\right] \geq 1 - 1/n\,.$$

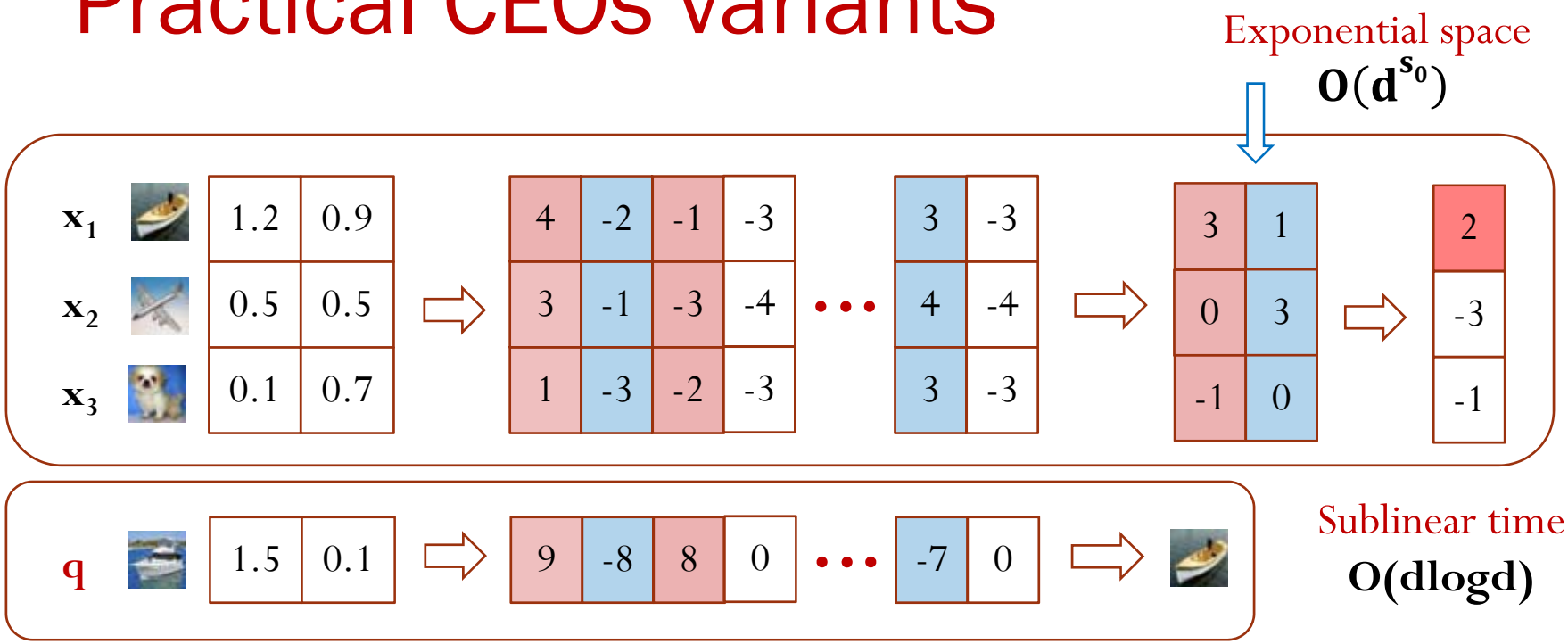
CEOs: Complexity

- Settings and conditions:
  - Given a constant $\mathbf{s_0}$, CEOs uses $2\mathbf{s_0}$ concomitants of extreme order statistics.
  - Assume $\boldsymbol{\alpha_*^2} = \mathbf{argmin}_{\mathbf{y}\in\mathbf{X}\setminus\mathbf{x}}\ \boldsymbol{\alpha_y^2} = \mathbf{c/s_0}$ and $\mathbf{c > 1}$.
  - CEOs uses $\mathbf{D} = \mathbf{n}^{1/c} = \mathbf{o(n)}$ random projections.
- Exponential indexing space:
  - Complexity: $\mathbf{O}\left((\mathbf{D/s0})^{\mathbf{s_0}}\right) = \mathbf{O}\left(\mathbf{n}^{1/c}\right) = \mathbf{O}\left(\mathbf{n}^{1/\alpha_*^2}\right)$
  - On unit sphere, to distinguish between $\mathbf{R}$-near neighbor and $(\mathbf{1} + \boldsymbol{\varepsilon})\mathbf{R}$-near neighbor, CEOs uses $\mathbf{O}\left(\mathbf{n}^{1/\varepsilon^2}\right)$ space, matching the lower bound [AIP06].
- Sublinear querying time:
  - $\mathbf{O(dD) = O(dn^{1/c})}$ if using standard Gaussian projections.
  - $\mathbf{O(DlogD) = O(n^{1/c}\,logn)}$ if using random rotation [AIL+15] to simulate Gaussian projections.

Practical CEOs variants

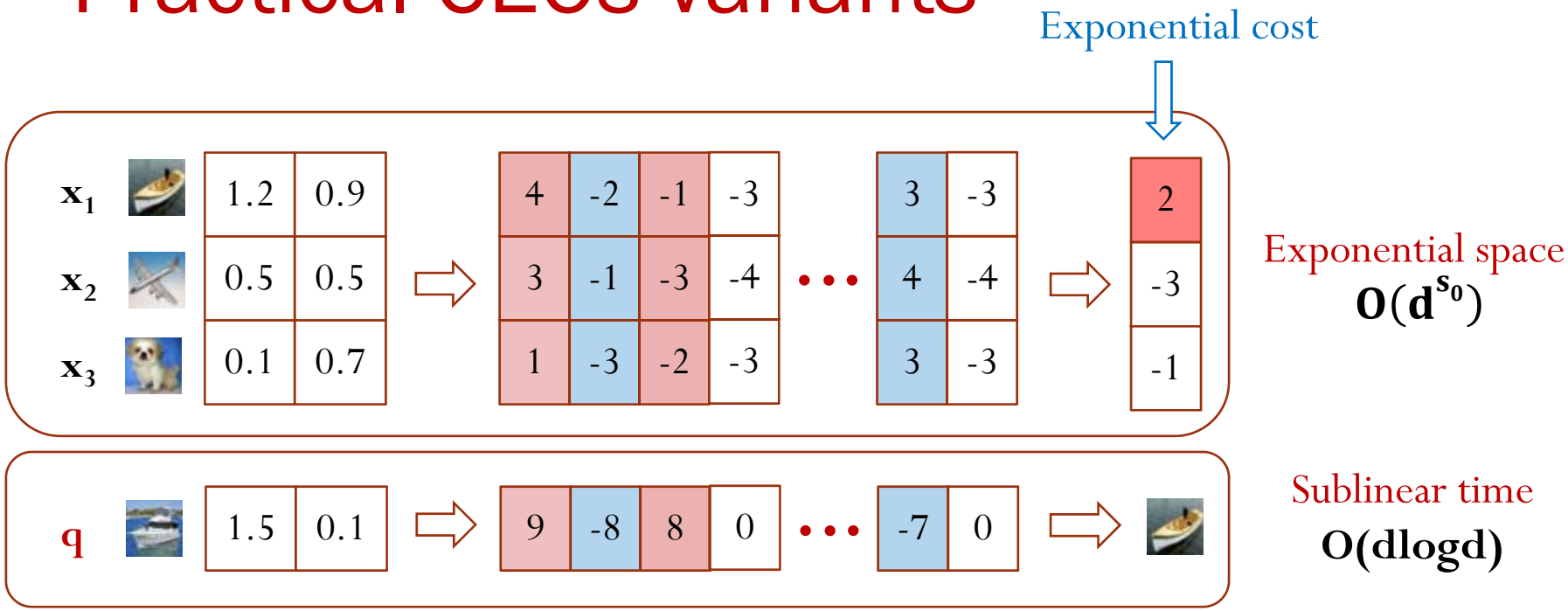


Practical CEOs variants



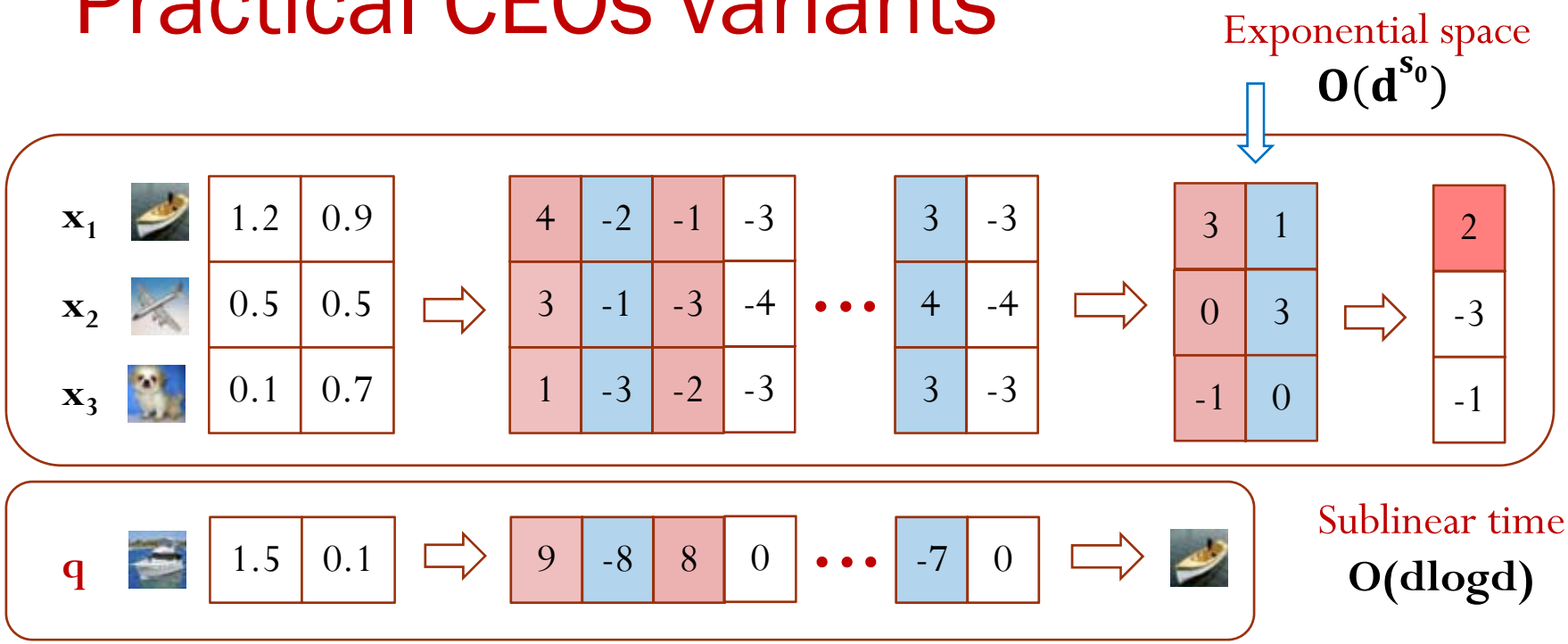
- Instantiating CEOs as sublinear MIPS solvers:
  - **D = O(d)** since we deal with high dimensional data sets.
  - 1CEOs uses 1 concomitant of the maximum order statistics.
  - 2CEOs uses 2 concomitants of the maximum and minimum order statistics.

Practical CEOs variants

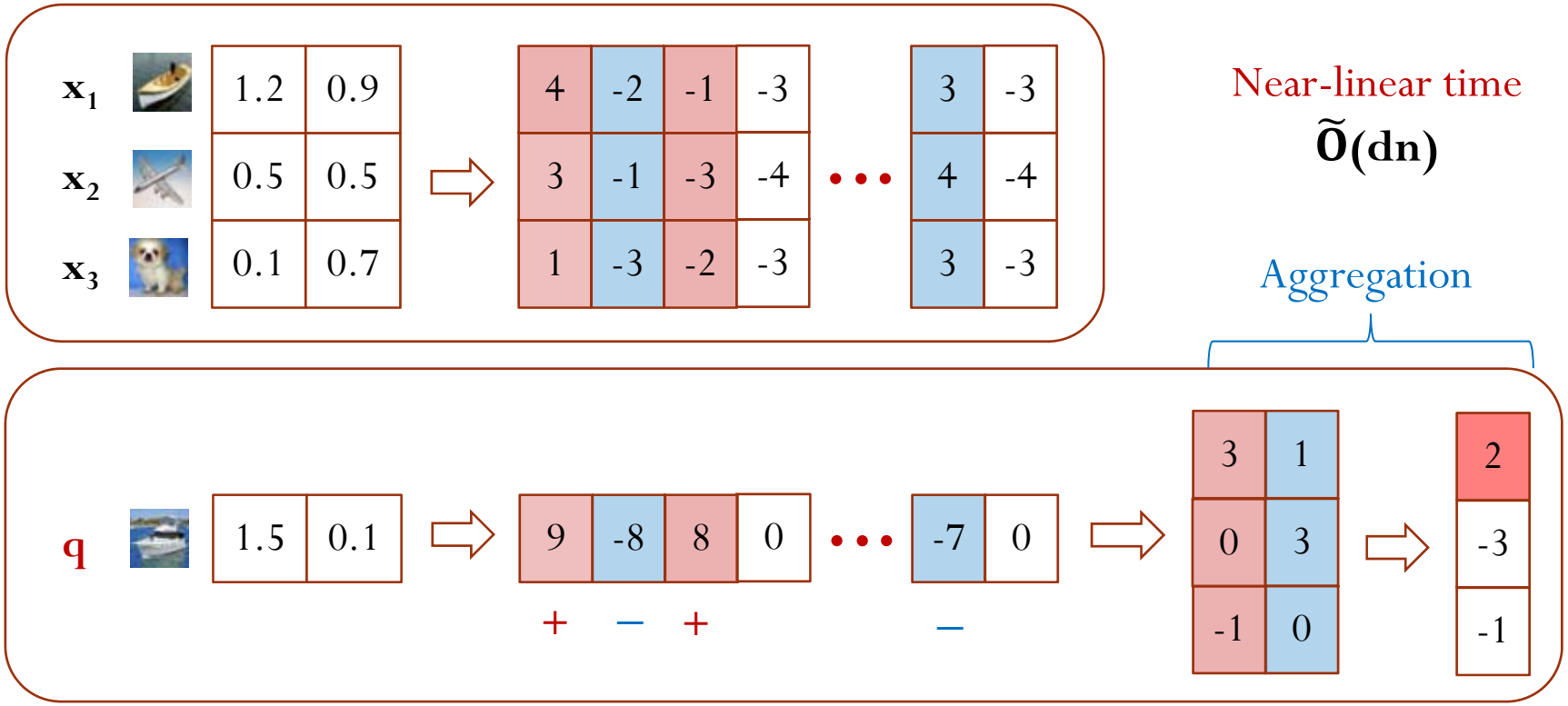


- Instantiating CEOs as sublinear MIPS solvers:
  - **D = O(d)** since we deal with high dimensional data sets.
  - 1CEOs uses 1 concomitant of the maximum order statistics.
  - 2CEOs uses 2 concomitants of the maximum and minimum order statistics.

Practical CEOs variants



Practical CEOs variants



- CEOs-Est aggregates to estimate all inner products in $\mathbf{O(n)}$ time.
- CEOs-TA exploits the threshold algorithm (TA) for top-k MIPS.
- coCEOs improves CEOs-TA by utilizing a larger $\mathbf{s_0}$ concomitants.

Experiment setups for top-10 MIPS

- Metrics:
  - We use a post-processing to compute exact inner product values of top-**b** candidates to return top-k MIPS

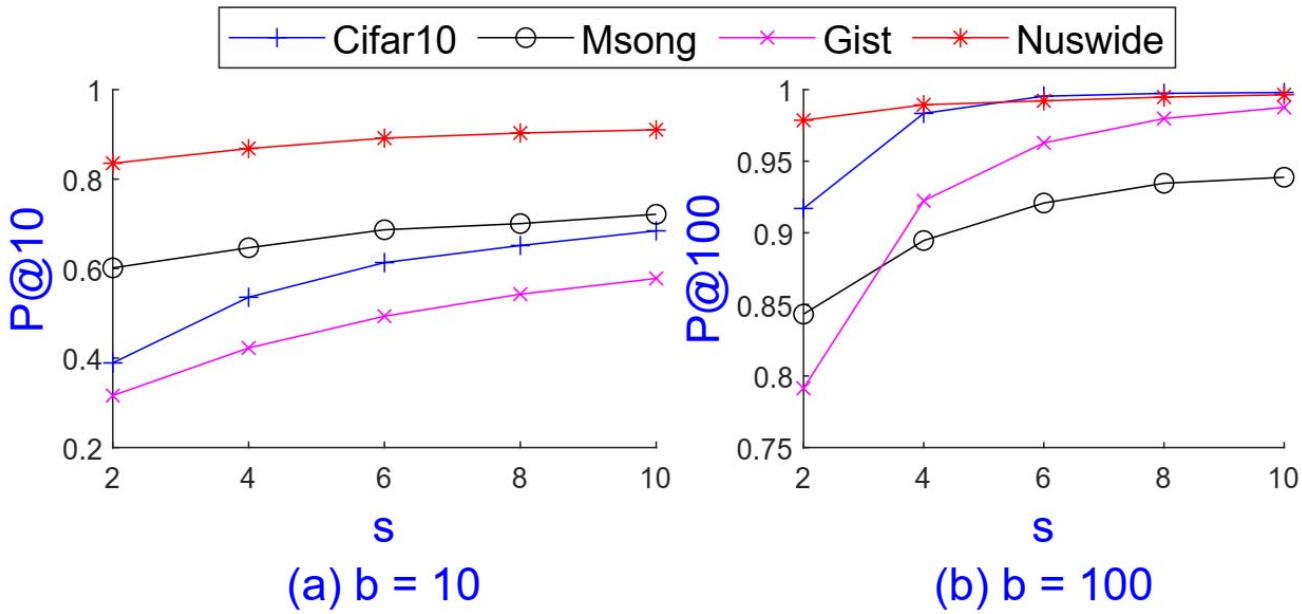
$$P@b = |\text{Retrieved top-}b \cap \text{True top-10}|/10$$

- Methods (github.com/NinhPham/MIPS):
  - LSH variants: SimHash [Cha02], SimpleLSH [NS15], RangeLSH [YLD+18]
  - Bruteforce with Eigen library C++
  - Delaunay graph: ipNSW [MB18] (see on the paper)
  - Budgeted MIPS: dWedge [LP20], Greedy [YHL+17] (see on the paper)

- Public datasets:
  - Cifar10, Nuswide, Yahoo, Msong, Gist, Imagenet, Tiny5m

CEOs search recall

- Parameter settings: $\mathbf{D = 2^{\lfloor \log d \rfloor + 1} = O(d),\ s = 2s_0}$



- Observation: CEOs uses $\mathbf{O(dn+d^2)}$ indexing space and $\mathbf{O(d\log d)}$ query time can achieve at least 80% search recall.

CEOs vs. LSH as sublinear solvers

- **(l, L)**-parameterized LSH bucket algorithms:
  - Fix **L = 512** hash tables and select the optimal **l** concatenating hash functions to return the top-100 candidates.

Algorithms	Index		Query	
	Space	Time	*P*@100	Time
1CEOs	**0.8 GB**	**1.3 mins**	51%	**2.3 ms**
2CEOs	7.5 GB	1.2 hours	**80%**	**2.3 ms**
SimpleLSH_{24}	9.1 GB	1.5 hours	35%	8.1 ms
RangeLSH_{16}	9.1 GB	1.1 hours	46%	5.9 ms

- On Gist: 1CEOs and 2CEOs outperform LSH variants in term of indexing and querying performance.

CEOs vs. LSH as MIPS estimation

- Hashing code length: $\mathbf{l = 128}$

Algorithms	Index		Query	
	Space	Time	$P@50$	Time
CEOs-Est	7.6 GB	72 s	**97%**	9 ms
CEOs-TA	14.7 GB	315 s	**97%**	**2.8 ms**
coCEOs	7.5 GB	176 s	**98%**	**2.2 ms**
SimHash_{128}	7.1 GB	30 s	89%	16.7 ms
SimpleLSH_{128}	7.1 GB	30 s	73%	11.3 ms
RangeLSH_{128}	7.1 GB	53 s	76%	15.4 ms

- On Gist: Among CEOs variants ($\mathbf{s_0 = 5}$), the inferior CEOs-Est outperforms LSH variants regarding querying performance.

Conclusion

- CEOs in theory:
  - CEOs has sublinear query time with search recall guarantees.
  - CEOs is an optimal dimensionality reduction on unit sphere.
- Light-weight indexing CEOs variants:
  - Simple to implement and efficient on indexing (just in minutes).
  - Competitive to recent MIPS solvers in term of querying performance.
- Open problem:
  - Is CEOs optimal on the whole space?

```python
import numpy as np

DATA = np.loadtxt("data.txt")  # Load data matrix N x D
(n, d) = DATA.shape  # N x D
D = 2**10  # Model parameter
RANDOM = np.random.normal(0, 1, (d, D))  # Generate N(0, 1)
DATA = np.matmul(DATA, RANDOM)  # Random projection on data
QUERY = np.loadtxt("query.txt")  # Load query matrix Q x D
QUERY = np.matmul(QUERY, RANDOM) # Random projection on query

for query in QUERY:
    minIdx = query.index(min(query)) # Minimum extreme order statistics
    maxIdx = query.index(max(query)) # Maximum extreme order statistics
    Est = DATA[maxIdx, :] - DATA[minIdx, :]
    print("Top1 MIPS is %f with value %f", Est.index(max(Est)), max(Est))
```